

Research Statement

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My research interests lie in quantum computing, circuit quantum electrodynamics, and quantum controls, with a particular focus on superconducting quantum devices and microwave pulse engineering. During the summers before my eighth semester, I completed an internship at the Bhabha Atomic Research Centre (BARC), where I worked on pulse-level control schemes and techniques under the supervision of Dr. Prashant Shukla¹ and Sanndeeep Joshi²

In this project, we did a comparative study of single-qubit gate operations in a transmon system inspired by Ref.[1]. Our study includes the calculation of average gate infidelity, analysis of leakage and seepage processes, comparison of leakage spectra, and phase error correction through virtual-Z gate implementation.

The system — Single-qubit rotations of superconducting qubits are typically implemented with nearly resonant microwave drive pulses with the drive pulses applied to a drive line, which is either capacitively or inductively coupled to the qubit. In the frame rotating at the microwave drive frequency $f_d = \omega_d/2\pi$, the Hamiltonian of a driven nonlinear oscillator Transmon[2] can be written as [1]

$$H_R = \sum_j \delta_j |j\rangle\langle j| + \frac{\Omega^x(t)}{2} \sigma_x^{j-1,j} + \frac{\Omega^y(t)}{2} \sigma_y^{j-1,j}. \quad (1)$$

where $\delta_j = \omega_j - j\omega_d$ is the detuning corresponding to the j -th energy level with energy ω_j , $\Omega^x(t)(I)$ and $\Omega^y(t)(Q)$ are the in-phase and quadrature components of the complex envelope $\Omega^{xy}(t) = \Omega^x(t) - i\Omega^y(t)$ related to the control pulse $\Omega^f(t) = \Omega^x(t) \cos(\omega_d t) + \Omega^y(t) \sin(\omega_d t)$ with a pulse duration t_p , and the Pauli-like operators are given by $\sigma_x^{j-1,j} = \lambda_j(|j\rangle\langle j-1| + |j-1\rangle\langle j|)$, $\sigma_y^{j-1,j} = i\lambda_j(|j\rangle\langle j-1| - |j-1\rangle\langle j|)$.

The transmon qubit is modeled as a three-level system, characterized by an anharmonicity of $\alpha/2\pi = 212$ MHz and a qubit frequency of $\omega_q/2\pi = 4.998$ GHz. In this framework, the in-phase component $I(t)$ and quadrature component $Q(t)$ of the drive pulse induce rotations around the x - and y -axes, respectively, within the computational subspace. The corresponding rotation angle is determined by the time integral of the pulse envelope. For our analysis, we have implemented single-qubit X -gate, with the pulse amplitude calibrated such that the rotation angle equals π .

Here we are applying an additional pulse in the y -quadrature for mitigating the leakage, popularly known as Derivative Removal Adiabatic Gate (DRAG) - pulse, which is ideally the derivative of in-phase pulse $\Omega^y(t) = -\beta \dot{\Omega}^x(t)/\alpha$. In Fig. 1, we illustrate the effect of four-different Gaussian DRAG [3] schemes on fidelity under time-dependent detuning $\delta(t)$, typically implemented via a flux line. Gate performance was quantified through average gate infidelity across multiple initial states. The colored vertical lines mark the gate durations required to reach an error of $\sim 10^{-2}$ for corresponding color pulse schemes. Notably, the

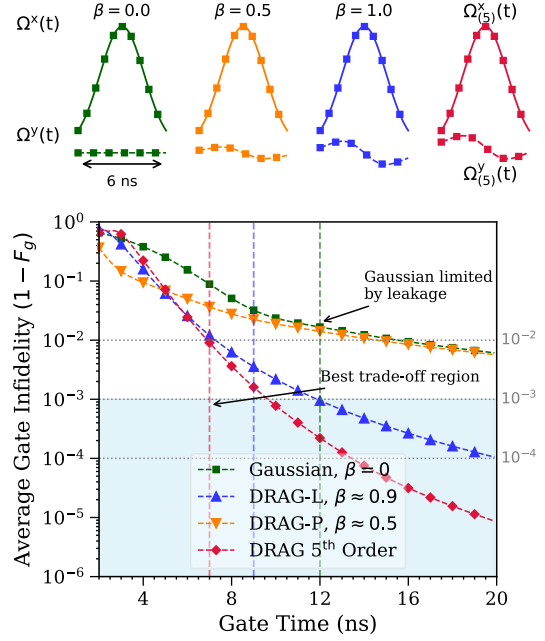


Figure 1: Pulse envelopes $\Omega^x(t)$ and $\Omega^y(t)$ for different DRAG parameters β , with $\beta = 0$ corresponding to a Gaussian pulse, $\beta = 0.5$ to the DRAG-P scheme (phase correction), $\beta = 0.9$ to the DRAG-L scheme (leakage suppression), and the 5th-order DRAG correction [3]. (Bottom) Average gate infidelity ($1 - \mathcal{F}$) versus gate time (t_g), showing leakage-limited performance of the Gaussian and improved fidelity with DRAG schemes in the best trade-off region.

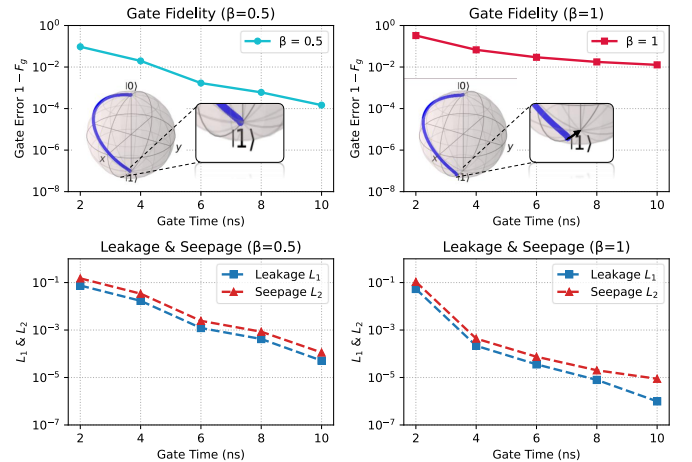


Figure 2: Gate performance for Gaussian DRAG pulses with $\beta = 0.5$ (left) and $\beta = 1.0$ (right). (Top) Gate fidelity versus gate time, with Bloch sphere trajectories showing qubit evolution. The $\beta = 0.5$ case (DRAG-P) provides phase correction with limited leakage suppression, while $\beta = 1.0$ (DRAG-L) yields stronger leakage suppression. (Bottom) Leakage $\mathcal{L}_1$ and seepage $\mathcal{L}_2$ [4] as functions of gate time, both decreasing with longer gates, with $\beta = 1.0$ giving superior suppression compared to $\beta = 0.5$.

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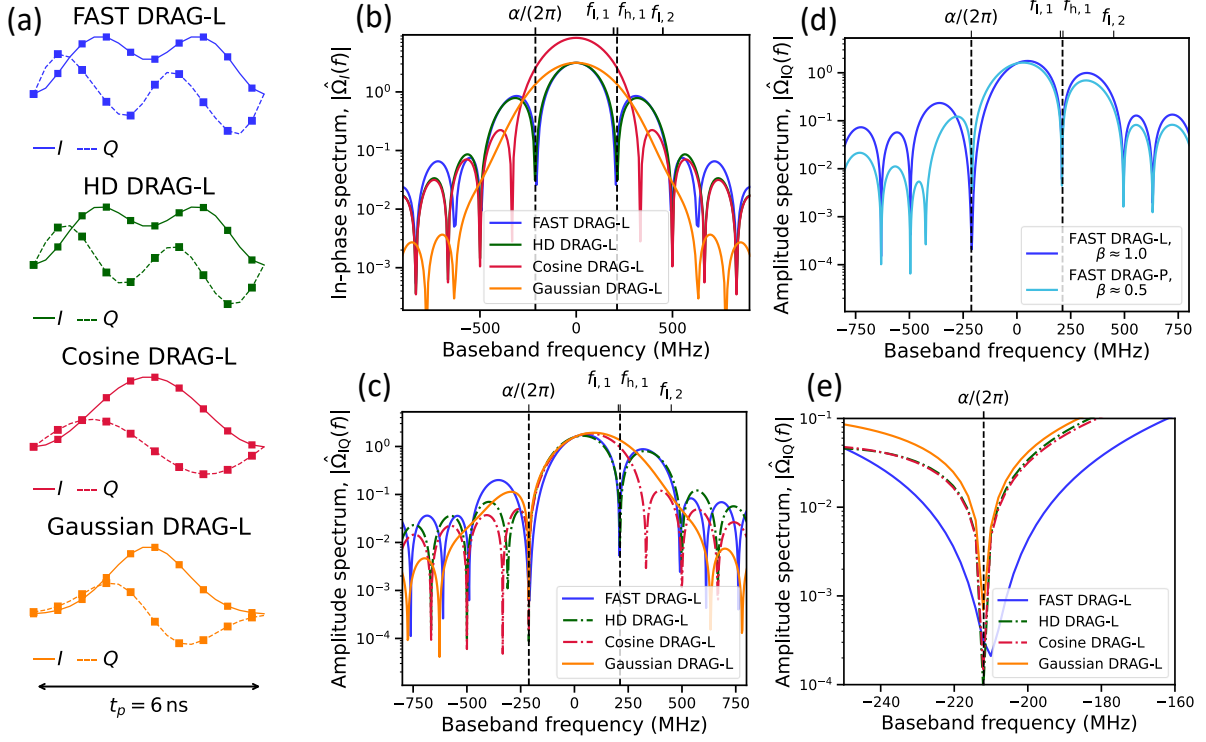


Figure 3: (a) Envelopes of the in-phase (I , solid) and quadrature (Q , dashed) components for two variants FAST DRAG-L (dark blue) and HD DRAG-L (darkgreen) with $\beta \approx 1.0$, both with same basis cosine function. The third one for conventional cosine DRAG pulse (red) with forth-one of extended gaussian DRAG pulse (dark orange) (b) Amplitude spectrum $|\hat{\Omega}_I(f)|$ of the in-phase envelope $I(t)$ for the same pulses. (c) Amplitude spectrum $|\hat{\Omega}_{IQ}(f)|$ of the complex IQ envelope $\Omega_{IQ}(t)$ for the same pulses, using DRAG to suppress the spectrum at the qubit anharmonicity $\alpha/(2\pi)$ (dashed line). (d) Leakage spectrum for the FAST DRAG-L and FAST DRAG-P variants shown maximum suppression by DRAG-L. (d) Zoomed-in plot of figure 3(c) shows the suppression comparisons of the above pulses.

5th-order DRAG and DRAG-L pulses both achieve 99% fidelity at ~ 7 ns, whereas Gaussian and DRAG-P pulses require about 12 ns to reach the same fidelity. Although higher-order DRAG and DRAG-L further suppress errors down to 10^{-4} , decoherence at longer gate times limits their advantage, yielding an optimal operating window (sweet point) between 7 and 12 ns.

In Fig. 2, we compare the gate infidelity ($1 - \mathcal{F}$) together with leakage ($\mathcal{L}_1$), of how much population leaked from the computational space and seepage ($\mathcal{L}_2$), of how much population comes back to the computational states. [4] as functions of gate time (t_g) for DRAG-P ($\beta = 0.5$) and DRAG-L ($\beta = 1.0$). For DRAG-P, the gate error decreases below 10^{-4} at final gate times, whereas for DRAG-L it saturates around 10^{-2} . Both schemes effectively suppress leakage: DRAG-P keeps it below 10^{-4} , while DRAG-L pushes it further below 10^{-6} . This comparison shows that fidelity is more sensitive to phase error accumulation evident from the Bloch sphere trajectories in the fidelity plots which is being corrected with time-dependent detuning in Fig. 1 than to leakage. Phase error generally being corrected both in the start and in the end of the pulse applied by applying Virtual-Z [5] gate at the software level.

Finally, we implemented Fourier Ansatz Spectrum Tuning (FAST DRAG) and Higher-Derivative DRAG (HD DRAG) [1], both of which enhance spectral control and suppress leakage at the anharmonic frequency. To implement fast, low-leakage single qubit gates, we combine the FAST method with DRAG by using FAST to shape

the in-phase envelope of the control pulse and applying DRAG to obtain the quadrature component to combine FAST DRAG illustrated in Fig 3(a). Its advantage over others include stronger and wider spectral suppression over the traditional raised cosine illustrated in Fig. 3(e). To achieve this, we apply the FAST method to suppress the spectrum of in-phase envelope across the frequency interval $\alpha/2\pi \in [f_{l,ef}, f_{h,ef}] = [f_{l1}, f_{h,1}]$ and use DRAG-L with $\beta \approx 1$ to further suppress the fourier transform $\hat{\Omega}_{IQ}(f)$ of the complex envelope. Along with this it also reduces the bandwidth of the control pulse or lower the peak amplitude by using a second frequency interval $[f_{l2}, f_{h,2}]$ to reduce the spectral energy above a given cutoff frequency $f_c = f_{l2}$. Both the FAST DRAG-L and FAST DRAG-P variants, the pulse and spectra are illustrated in Fig. 3(a) and Fig. 3(d). HD DRAG minimizes the in-phase envelope spectrum around the qubit anharmonicity while additionally using DRAG to further suppressing leakage. Here we consider derivative upto 2nd order in $\Omega_I(t)$ envelope and till 3rd order in $\Omega_Q(t)$.

Despite being similar, HD DRAG has less parameters to calibrate than FAST DRAG and can suppress multiple leakage transition by introducing higher derivatives. Although, HD DRAG does not allow straightforward control over the pulse bandwidth, the average power, or the exact strength and width of the spectral suppression unlike FAST DRAG, which offers more flexibility for the pulse design. Hence, we foresee that the flexibility of FAST DRAG is beneficial especially in scenarios with multiple leakage transitions.

The Conclusion — This work highlights various techniques for improving the overall fidelity of single-qubit gates, primarily by suppressing leakage errors. We have discussed several control schemes, including DRAG, fifth-order DRAG, FAST DRAG, and the HD DRAG method. Among these, the FAST DRAG scheme emerges as the most robust and flexible, with the added advantage of being easily calibratable at the hardware level. Additionally, we examined the sensitivity of average gate fidelity to phase errors. These errors can be effectively mitigated using software-level corrections such as the virtual-Z gate, applied before and after the control pulse. Alternatively, phase accumulation can be minimized through time-dependent detuning, which dynamically adjusts the qubit energy levels during gate operation.

Future-aspects – We can develop a mechanism to operate on the intermediate states of a multi-qubit system by designing pulses that first suppress spectral components at the primary anharmonicity, followed by suppression at the secondary anharmonicity.

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